

Murthal Parantha

The coding blocks members went to the success party of their first ever online boot-camp at Murthal. They ordered P number of paranthas. The stall has L cooks and each cook has a rank R . A cook with a rank R can cook 1 parantha in the first R minutes, 1 more parantha in the next $2R$ minutes, 1 more parantha in $3R$ minutes and so on (he can only cook a complete parantha). (For example if a cook is ranked 2, he will cook one parantha in 2 minutes one more parantha in the next 4 mins and one more in the next 6 minutes hence in total 12 minutes he cooks 3 paranthas. In 13 minutes also he can cook only 3 paranthas as he does not have enough time for the 4th parantha). Calculate the minimum time needed to cook all the paranthas.

Input Format

First line contains P , the number of pratha ordered. In the next line the first integer denotes the number of cooks L and L integers follow in the Next line each denoting the rank of a cook.

Constraints

$P \leq 1000$
 $L \leq 50$
 $1 \leq R \leq 8$

Output Format

Print an integer which tells the number of minutes needed to get the order done.

Sample Input

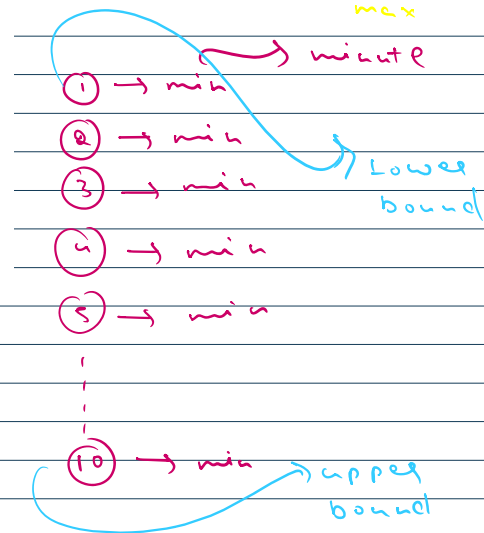
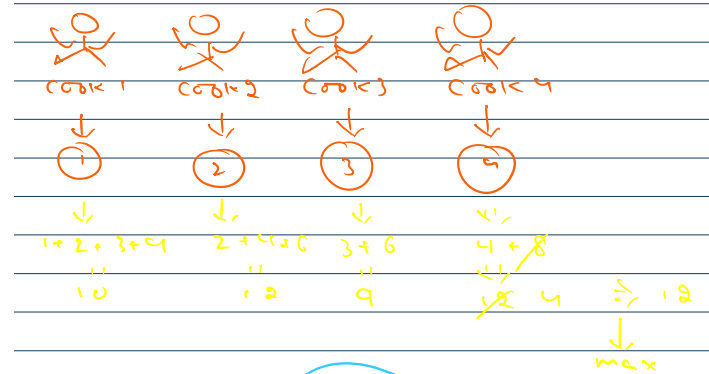
10 $\rightarrow P \Rightarrow$ no of paranthas
 4 $\rightarrow L \Rightarrow$ no of cooks
 1 2 3 4 $\Rightarrow$ Rank[]

Sample Output

12

Explanation

First cook with rank 1 cooks 4 paranthas in 10 minutes ($1+2+3+4$).
 Second cook with rank 2 cooks 3 paranthas in 12 minutes ($2+4+6$).
 Third cook with rank 3 cooks 2 paranthas in 9 minutes ($3+6$).
 Fourth cook with rank 4 only needs to cook one last remaining parantha. He can do that in 4 minutes.
 Since these cooks cook parallelly, the total time taken will be the maximum of the four i.e. 12 minutes.



Find Square Root

$n = 28$ $\rightarrow$ floor value

$n = 36$ $\rightarrow$ exact value

1st part: $low = mid - 1$
 2nd part: $low = mid + 1$

$l = 1$ $\rightarrow$ $m = 18$
 $h = 36$

$18 * 18 \geq 36$ ✓

$\rightarrow$ 1st part

$9 * 9 \geq 36$ ✓

$\rightarrow$ 1st part

$4 * 4 \geq 36$

$\rightarrow$ 2nd part

$l = 5$
 $h = 8$ $\rightarrow$ $mid = 6$ $0 * 6 == 36 \rightarrow$ Answer